

Classification

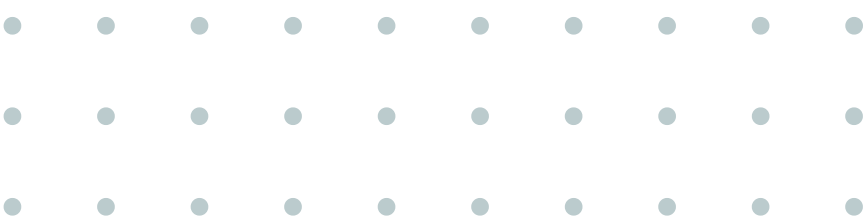
Introduction to Predicting Categories with ML
By Muhammad Nanda S. / Data Scientist



Session Objectives

01. Understand **the classification concept**

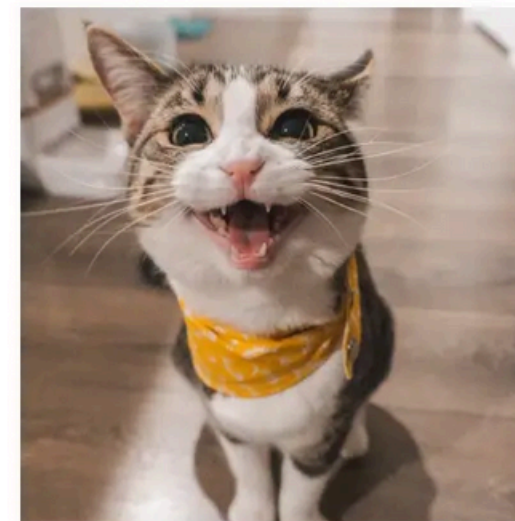
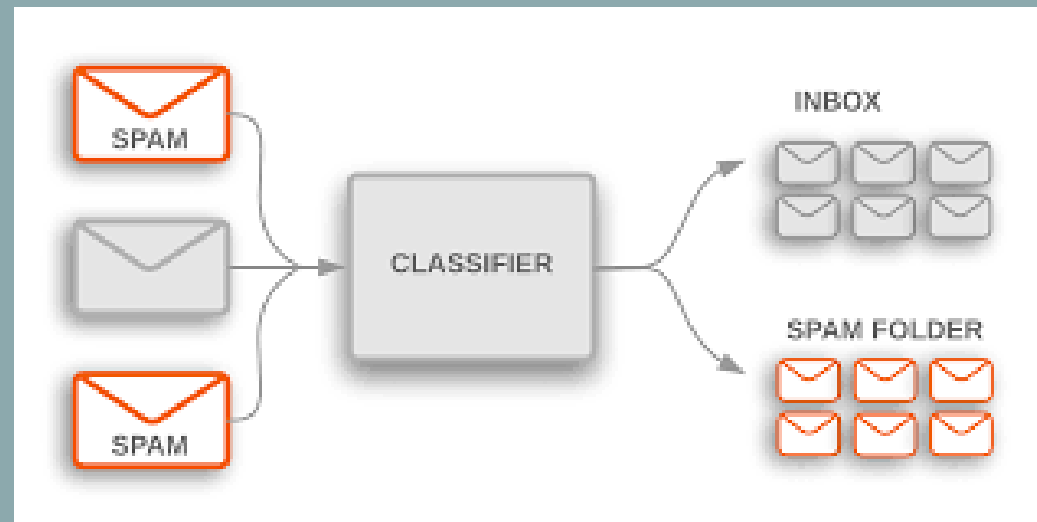
02. Understand **how to evaluate** classification models performance



What is Classification?

Classification is a type of supervised learning where the goal is to predict a **discrete label**.

- It's about answering "Which class?" or "Is it X or Y?"
- The output is a category (e.g., Spam / Not Spam, Cat / Dog, Disease / Healthy).



Classification
Cat

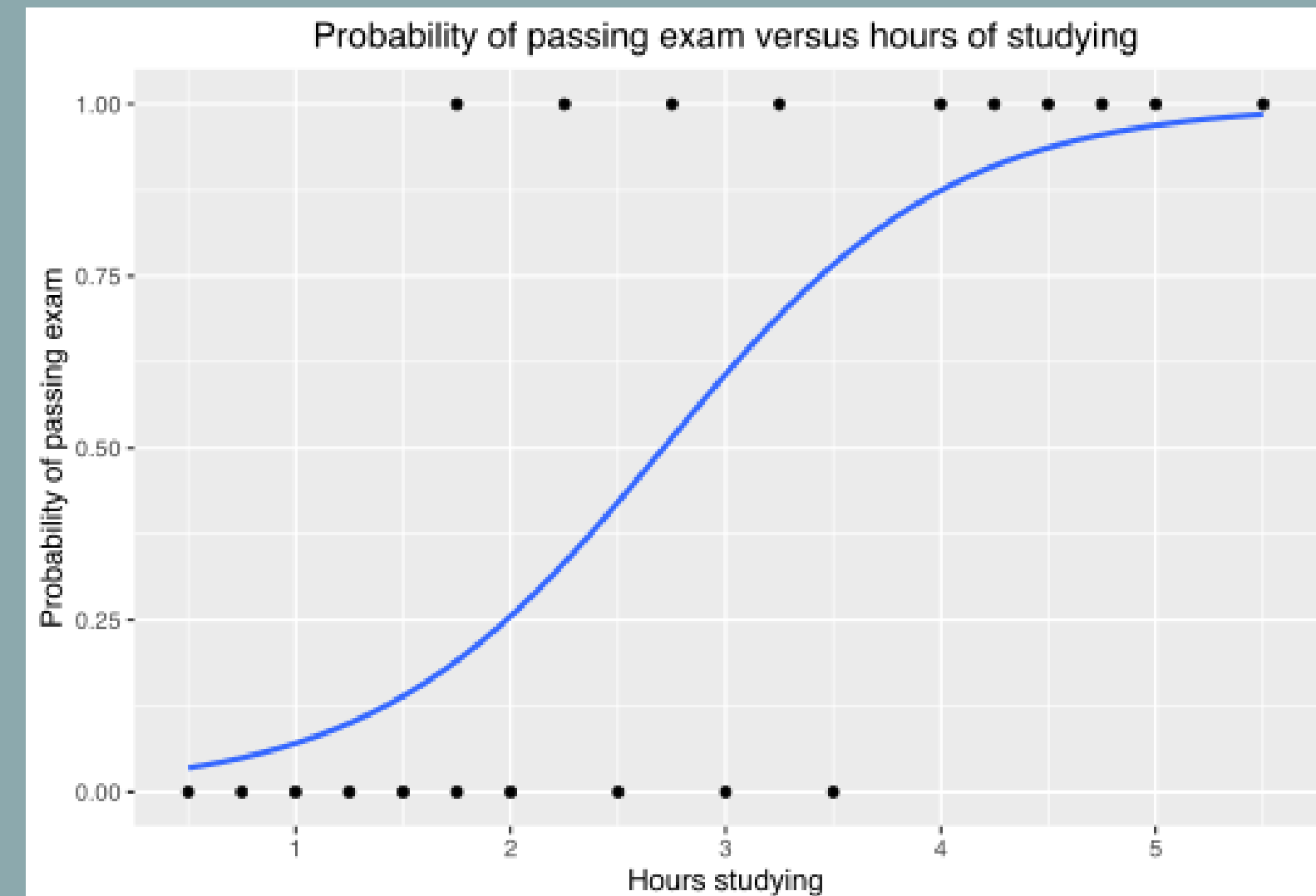


Classification use case: spam filtering, image classification, disease detection



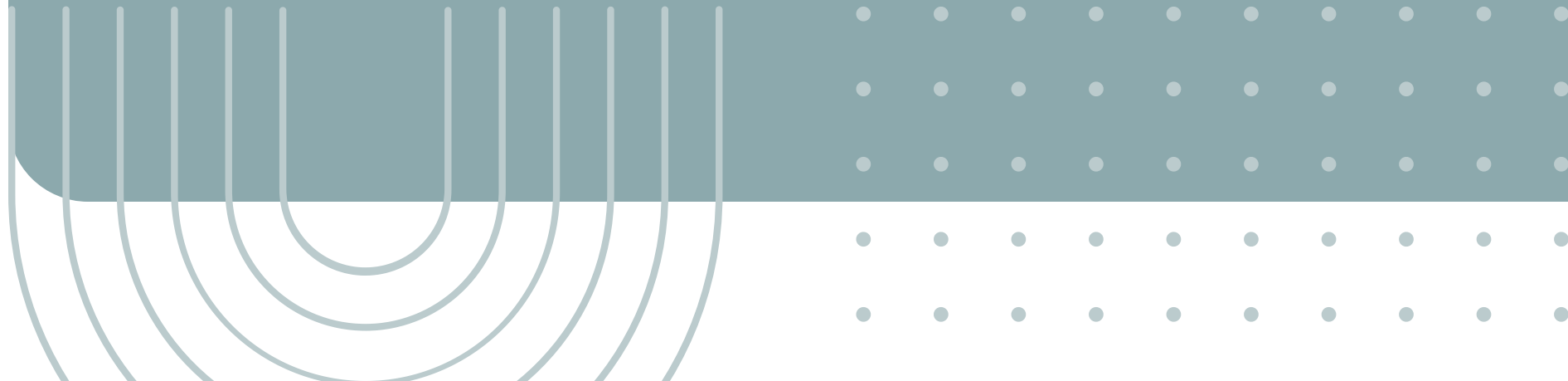
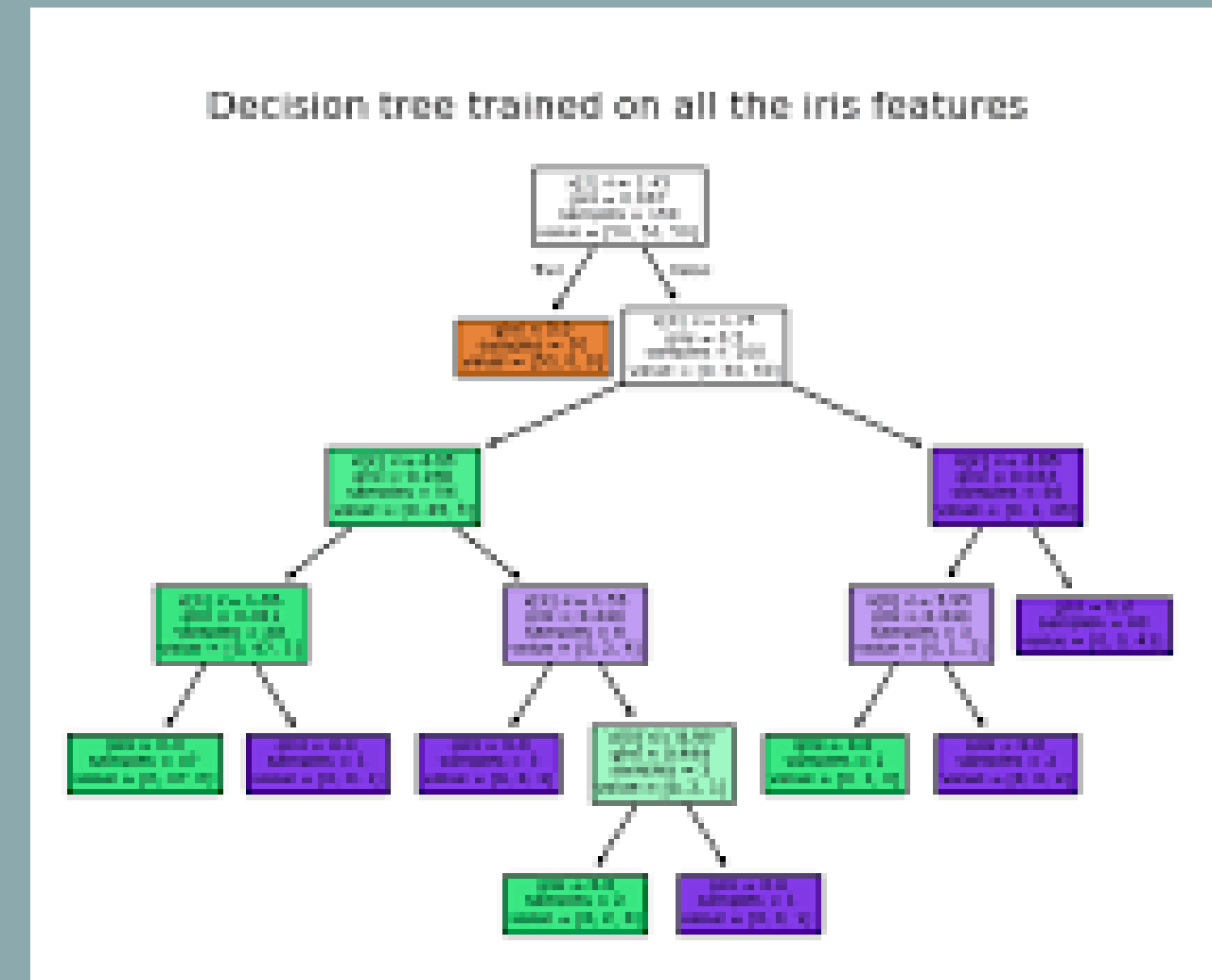
The Basic: Logistic Regression

- Finds the best-fit **sigmoid curve** to predict the **probability of a binary outcome**.
- Pros/Cons:
 - + : Interpretable and works well when classes are linearly separable.
 - - : Struggles with complex relationships or non-linear boundaries.
- Use when the target variable is binary (e.g., yes/no, 0/1) and a probabilistic interpretation is desired.



Decision Trees: Asking Questions to Predict

- **Splits the data into branches** based on feature values to make predictions
- **Pros/Cons:**
 - + : Easy to interpret, handles non-linear data well.
 - - : Prone to overfitting without pruning or constraints.
- Use when interpretability is important, or when data has complex, non-linear patterns.



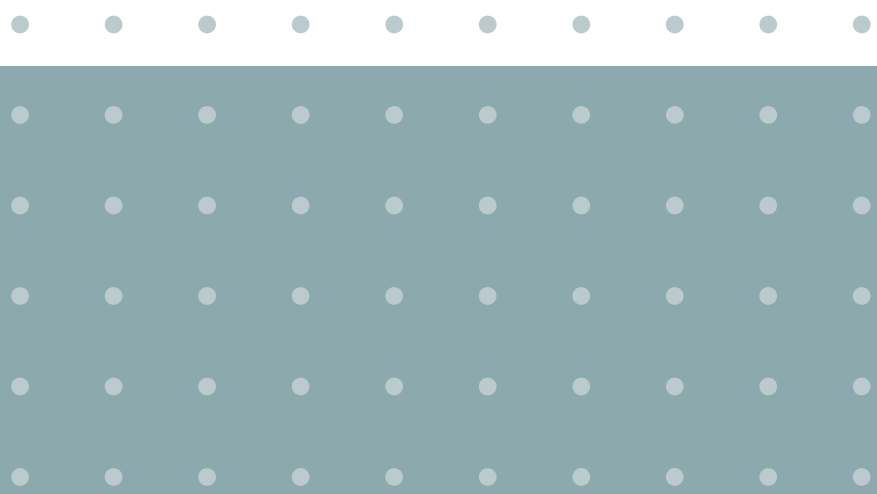
How Do We Measure Classifier Performance?

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Predictions}} = \frac{TP + TN}{TP + TN + FP + FN}$$

- Idea: Measures the proportion of correct predictions (true positives and true negatives) out of all predictions.
- Use/Effect: Works well when classes are balanced; misleading with imbalanced datasets where one class dominates.

		Predicted Values	
		Positive	Negative
Actual Values	Positive	TP	FN
	Negative	FP	TN

Term	Meaning	In a Simple Way	Example
True Positive (TP)	Model predicted positive, and it was actually positive.	✔ Got it right when it said “Yes”	Spam → Predicted Spam (Caught spam correctly)
False Positive (FP)	Model predicted positive, but it was actually negative.	✗ Said “Yes” but it was “No”	Not Spam → Predicted Spam (Marked a real email as spam)
False Negative (FN)	Model predicted negative, but it was actually positive.	✗ Said “No” but it was “Yes”	Spam → Predicted Not Spam (Missed a spam email)
True Negative (TN)	Model predicted negative, and it was actually negative.	✔ Got it right when it said “No”	Not Spam → Predicted Not Spam (Correctly let a real email through)



Let's Discuss!

Do you have any question?



Is this classification?

